

Signal Processing

MENG 4050

REZA MADJLESI

Intro to the Fundamentals of Computer Engineering

01. Intro to Computer Engineering
02. Digital Logic and Boolean Algebra
03. Computer Architecture
04. Programming Fundamentals
05. Data Structure and Algorithms

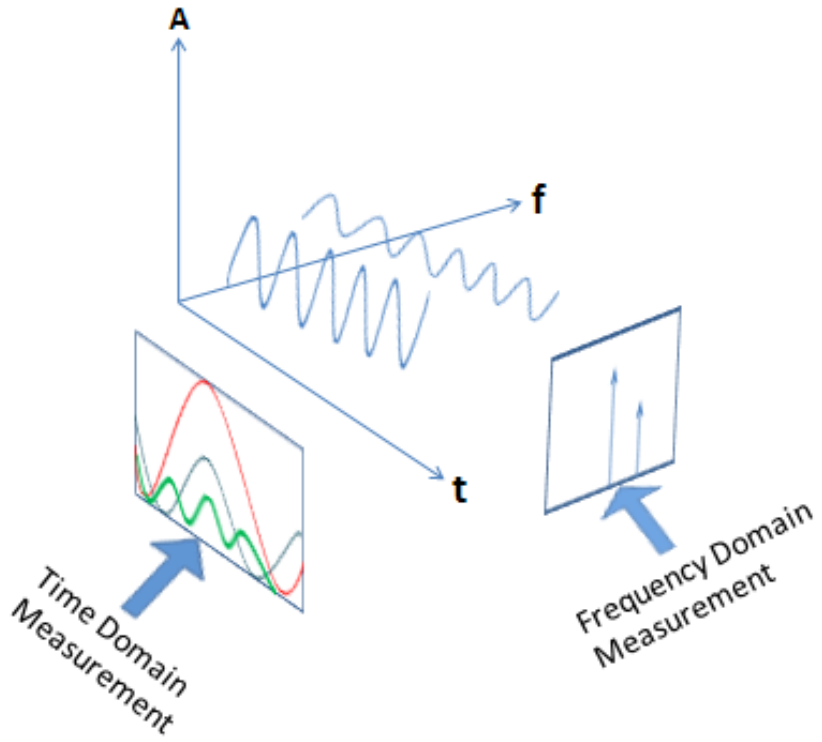


Signal Processing and Time Data

Time domain signal:

- The raw signal (amplitude versus time) from any sensor
- collected signal through the DAQ device
- Time data analyzed to get the required information.

There are several signal processing techniques are available to analyze the measured the data to get the meaningful information from the signals





Section One

We will cover Filter

- Basic concepts
- Overview of hardware and software
- Intro to computer systems and application

Filters

Low pass (LP) filter:

it is generally applied on the recorded/collected/measured data. It removes all frequencies within the measured signal at and above the frequency (fl). The user can choose the value of the low pass frequency (fl) depending on the requirements.

High pass (HP) filter:

In many cases it is required to filter out low frequency noise, DC components, etc. from the recorded/ measured signals before further signal processing.

Band pass (BP) filter:

This is a combination of both HP and LP filters.

Filters in Signal Processing

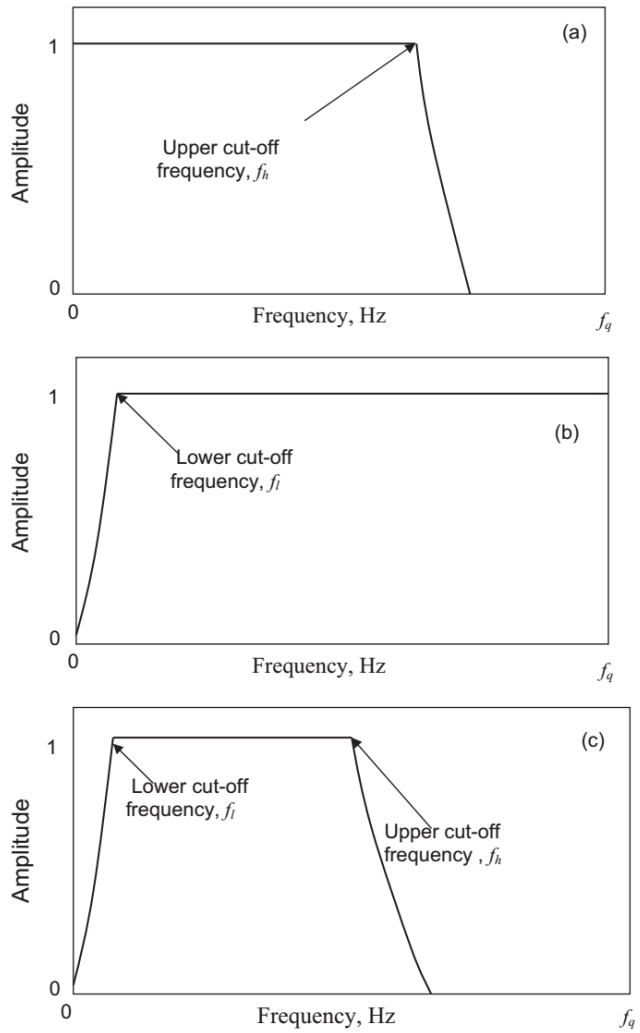


FIGURE 5.1 (a) Low-pass filter, (b) high-pass filter, (c) band-pass filter characteristics.

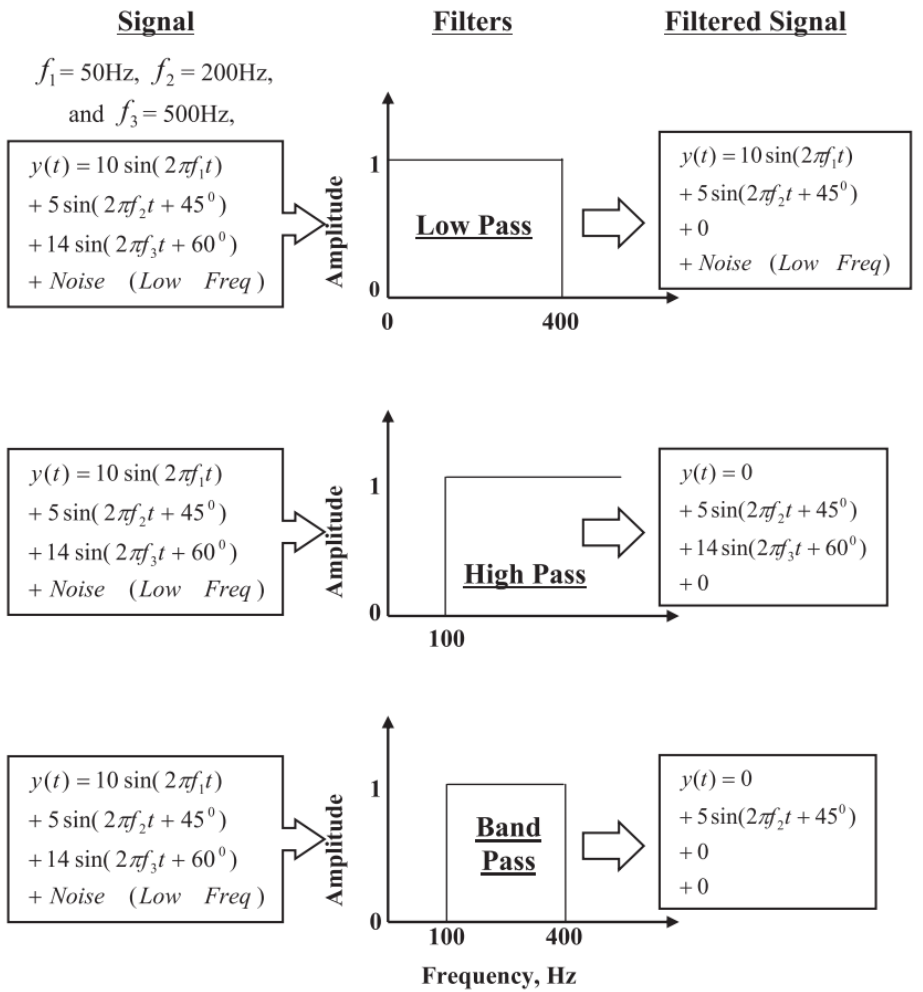


FIGURE 5.2 Working illustrations of different filters.



Section Two

We will cover Statistical Analysis

- Basic concepts
- Overview of hardware and software
- Intro to computer systems and application

Amplitude of Vibration

The overall amplitude of any signal may be specified in any of three ways:

1. root mean square (RMS)
2. 0 to peak (pk) or
3. peak-to-peak (pk-pk)

wave signal at 50 Hz. Figure 5.3b however shows the time waveform of a periodic signal that is non-sinusoidal.

$$y_{rms} = \sqrt{\frac{\sum_{i=1}^{p} y_i^2}{p}}$$

The RMS value of any sine wave is always equal to 0.707 times the pk value.

If the measured signals may not be pure sinusoidal waveform it is better to estimation the RMS value numerically.

Amplitude of Vibration Representation

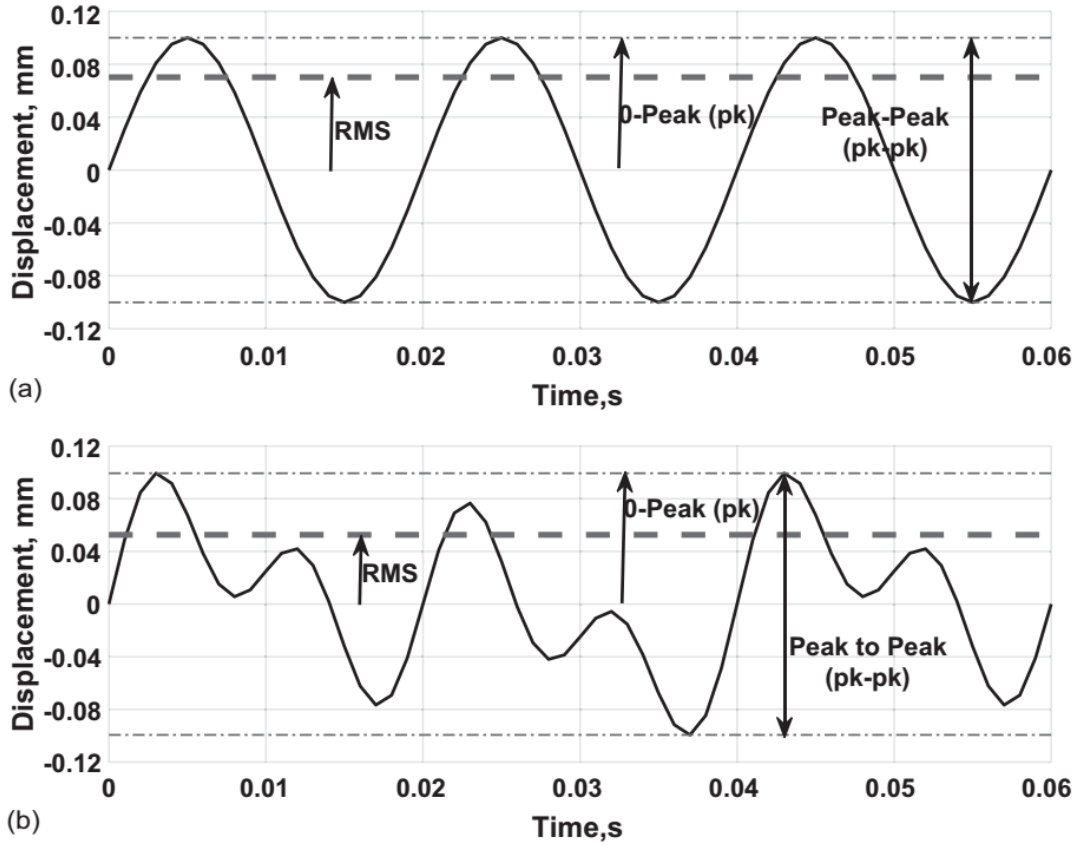


FIGURE 5.3 Time wave forms showing 0-peak, peak-peak, and RMS vibration amplitudes.

Integration of Time Domain Signal

Integration of Time domain Signal

The velocity at the time $t_i, v_i = \dot{y}_i = \ddot{y}_i dt + \dot{y}_{i-1}$

the displacement at the time $t_i, y_i = \dot{y}_i dt + y_{i-1}$

The displacement, $x(t) = 20\sin(3.5\pi t) + 25\cos(6\pi t)$

The velocity, $v(t) = 70\pi\cos(3.5\pi t) - 150\pi\sin(6\pi t)$

the acceleration, $a(t) = -245\pi^2\sin(3.5\pi t) - 900\pi^2\cos(6\pi t)$

Integration of Time Domain Signal, No filtering

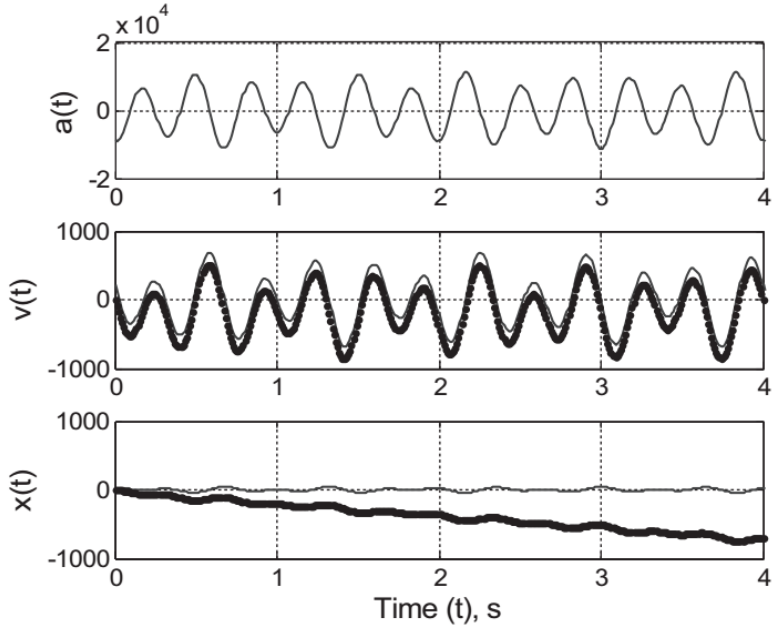


FIGURE 5.4 Original acceleration, velocity and displacement signals (solid line) and comparison with velocity and displacement signals (solid line with dots) computed from the original acceleration signals using the numerical integration.

Fixes to Integration of Time Domain

that the signals obtained from the numerical integration method show significant deviation from the original signals due to the fact that

the initial conditions are generally not known

that results into DC offset in the integrated signals due to the integration constant.

These DC components in the integrated signals (velocity and displacement) can be considered as the very low frequency components in the signal.

it is always recommended that the integrated signals must be high-pass filtered at a very low frequency.

Here the integrated signals—velocity and displacement signals—are high-pass filtered at 1 Hz. It can be seen that the filtered signals shown in Figure 5.5 are almost identical to the original signals after 0.5 s. Thus,

it is also recommended to remove some initial data after integration to get the reliable data. For the present example, the data up to 0.5 s must be removed from the integrated data sets.

Integration of Time Domain Signal, HP Filtered

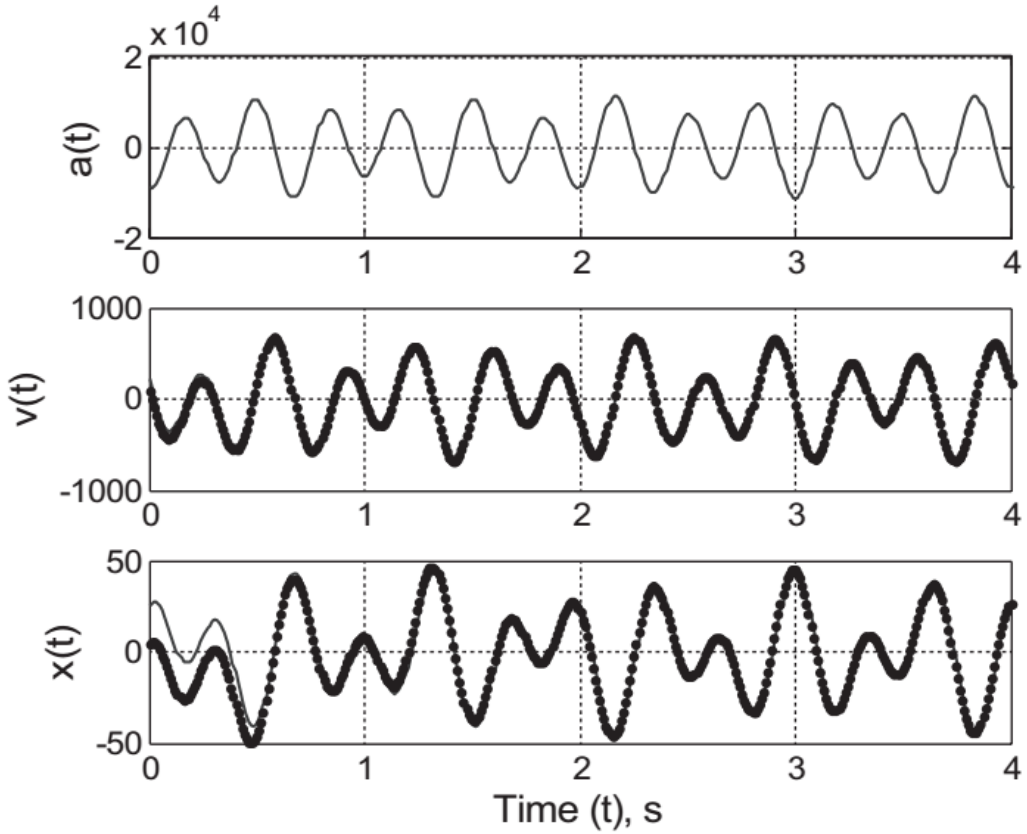


FIGURE 5.5 Original and integrated signals—velocity and displacement after high pass filter at 1 Hz showing promising results.

Statistical properties of Time Signals

Nature of the signal whether it is sine, random or impulsive.

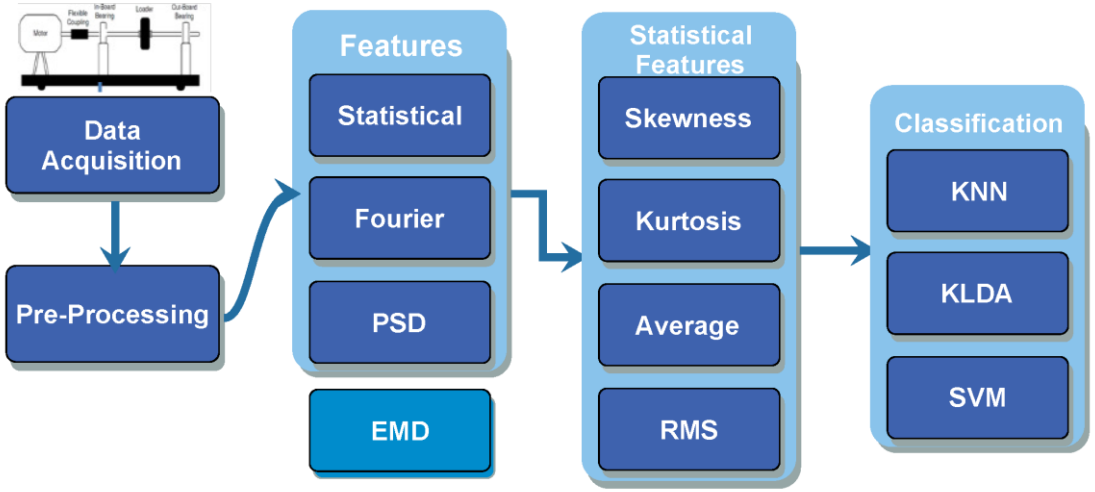
Crest Factor (CF):

1. a non-dimensional parameter for signal,
2. defined as the ratio of the peak value to the RMS value.
3. For sine wave, it is always going to be equal to 1.414 or 2 because yrms ypeak = 2 .

Kurtosis (Ku):

normalized fourth order moment of the time domain signal.

The definition of the kth order central moment for any time domain signal (say, y t()) is given by



$$Ku = \frac{M_4}{(M_2)^2}$$

$$M_k = \frac{1}{p}\sum_{i=1}^{p}(y_i - \bar{y})^k$$

Statistical properties of Time Signals

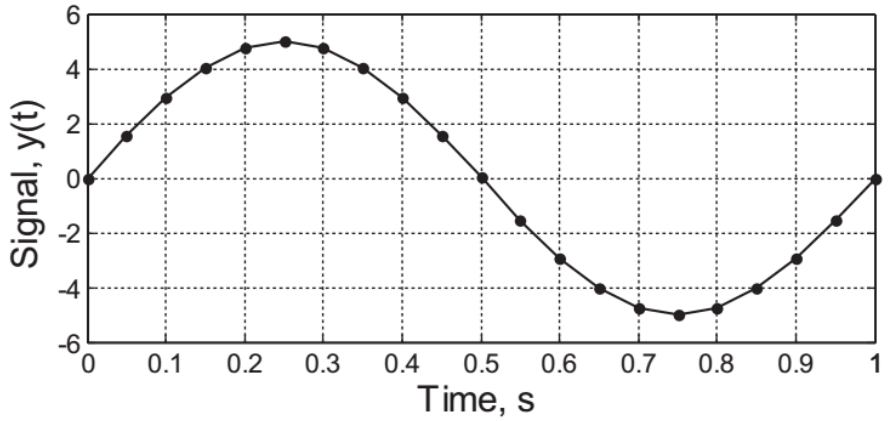


FIGURE 5.6 A typical signal, $y(t)$.

TABLE 5.1
Calculation of CF and Ku for a Sine Signal

S. No.	t, s	$y(t)$ or y_i		y_i^2	$(y_i - \bar{y})^2$	$(y_i - \bar{y})^4$
1	0	y_1	0.0	0.0	0.0	0.0
2	0.05	y_2	1.5451	2.3873	2.3873	5.6991
3	0.10	y_3	2.9389	8.6373	8.6373	74.6027
4	0.15	y_4	4.0451	16.3627	16.3627	267.7384
5	0.20	y_5	4.7553	22.6127	22.6127	511.3348
6	0.25	y_6	5.0000	25.0000	25.0000	625.0000
7	0.30	y_7	4.7553	22.6127	22.6127	511.3348
8	0.35	y_8	4.0451	16.3627	16.3627	267.7384
9	0.40	y_9	2.9389	8.6373	8.6373	74.6027
10	0.45	y_{10}	1.5451	2.3873	2.3873	5.6991
11	0.50	y_{11}	0.0	0.0	0.0	0.0
12	0.55	y_{12}	−1.5451	2.3873	2.3873	5.6991
13	0.60	y_{13}	−2.9389	8.6373	8.6373	74.6027
14	0.65	y_{14}	−4.0451	16.3627	16.3627	267.7384
15	0.70	y_{15}	−4.7553	22.6127	22.6127	511.3348
16	0.75	y_{16}	−5.0000	25.0000	25.0000	625.0000
17	0.80	y_{17}	−4.7553	22.6127	22.6127	511.3348
18	0.85	y_{18}	−4.0451	16.3627	16.3627	267.7384
19	0.90	y_{19}	−2.9389	8.6373	8.6373	74.6027
20	0.95	y_{20}	−1.5451	2.3873	2.3873	5.6991
21	1.00	y_{21}	0.0	0.0	0.0	0.0
			Mean, $\bar{y} = 0$	Sum, $\sum_{i=1}^{21} y_i^2 = 250$	$\sum_{i=1}^{21}(y_i - \bar{y})^2 = 250$	$\sum_{i=1}^{21}(y_i - \bar{y})^4 = 4687.50$
			Peak = 5	$y_{RMS} = \sqrt{\frac{\sum_{i=1}^{21} y_i^2}{21}} = 3.4503$	$M_2 = \frac{\sum_{i=1}^{21}(y_i - \bar{y})^2}{21} = 11.9048$	$M_4 = \frac{\sum_{i=1}^{21}(y_i - \bar{y})^{42}}{21} = 223.2143$

Crest factor, $CF = \frac{y_{peak}}{y_{rms}}$ **=1.4491**

Kurtosis, $ku = \frac{M_4}{(M_2)^2}$ **=1.575**

Kurtosis and Crest factor

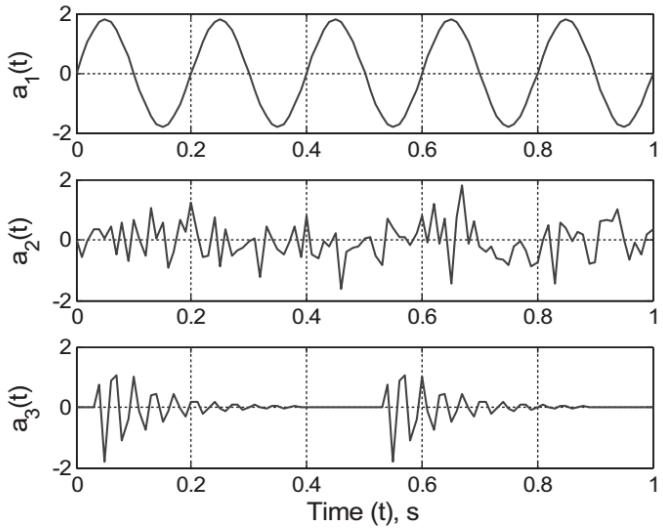


FIGURE 5.9 Three signal waveforms—sine, random, and transient (impulse response type).

TABLE 5.2
Comparison of Different Estimated Parameters for the Three Signals—Sine, Random and Impulsive (Transient)

Signals	Mean	Peak	RMS	CF	Ku
Sine, $a_1(t)$	1.892e-16	1.80	1.2728	1.4142	1.515
Random, $a_2(t)$	−3.851e-2	1.80	0.6108	2.9469	3.2563
Impulsive, $a_3(t)$	2.737e-5	1.80	0.4400	4.0905	8.4486

kurtosis

a better and reliable parameter to understand the nature of signals as it involves all data points within the signal for the computation of the kurtosis.

Crest factor

is simple but significantly dependent on a just one peak value of the signal. If the peak data in the signal is due to some measurement error then the CF value is going to be meaningless.



Section Three

We will cover Fourier Transform

- Basic concepts
- Overview of hardware and software
- Intro to computer systems and application

Fourier Transformation

The amplitude vs. frequency plot of any signal is known as the spectrum of the signal.

- define the time period (T)
- Frequency resolution, df = 1/T.
- Maximum frequency up to Nyquist frequency, fq

it is better to consider the spectrum up to the useful frequency limit, $f_u = \frac{f_s}{2.56}$ due the anti-aliasing filter used in the data collection.

Fourier transformation (FT) is a tool that is used for converting digital data in a time domain into the frequency domain. The process is also known as the Digital FT (DFT).

$$Y(kdf) = \frac{2}{N}\sum_{p=1}^{N} y_{(t_p)}\, e^{-j\frac{2\pi(p-1)k}{N}}$$

Fourier Transformation

Where, $Y(kdf) = Y(f_k)$is the FT at the frequency, $f_k = kdf$ for the time domain signal, y(t).

$Y_p = Y(t_p)$ is the amplitude of the signal, Y_t at time, $t_p = (p-1)dt$. N is the number of data points of the signal, Y_t used for the FT.

The variables k = 0, 1, 2,…, $(\frac{N}{2} - 1)$ and p = 1, 2, … , N. The frequency resolution, df = 1/T , where T is the pseudo (artificial) time period chosen for the signal for the FT and is estimated as T = Ndt, where dt= $\frac{1}{f_s}$ is the time between two consecutive data in the signal, y(t).

The number of data points (N) used for the FT calculation should be equal to 2^n, where n is a positive integer number.

The computed FT value, y(f_k) at the frequency, f_k is a complex quantity (i.e., ak + jbk).

Fourier Parameters

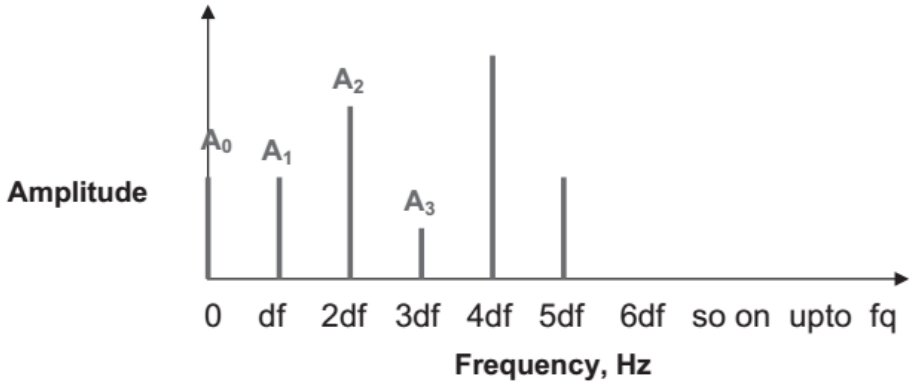


FIGURE 5.13 Spectrum plot: a graphical representation of the computed amplitude and frequency through FT using Equation (5.10).

TABLE 5.3
Parameters for FT Analysis

n	$N = 2^n$	$T = Ndt$, s	$df = \frac{1}{T}$, Hz	$f_u = \frac{f_s}{2.56}$, Hz	$N_L = \frac{N}{2.56}$
1	2	0.001953	512	400	0
2	4	0.003906	256	400	1
3	8	0.007812	128	400	3
4	16	0.015625	64	400	6
5	32	0.03125	32	400	12
6	64	0.0625	16	400	25
7	128	0.125	8	400	50
8	256	0.25	4	400	100
9	512	0.50	2	400	200
10	1024	1.0	1	400	400

Fourier Resolution

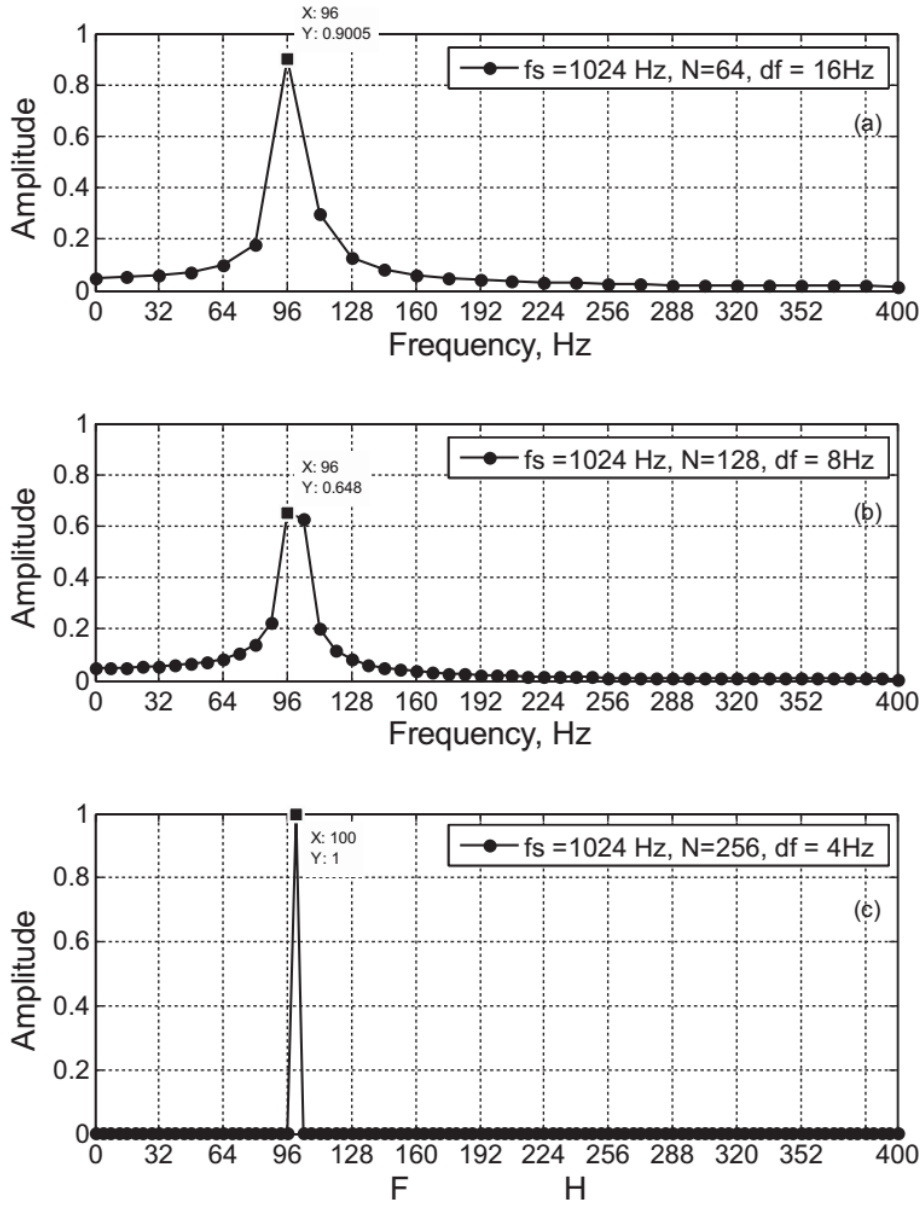


FIGURE 5.14 Estimated spectrum plots using the FT approach for the sine wave signal in Figure 5.10, (a) $N = 64$ and $df = 16$ Hz, (b) $N = 128$ and $df = 8$ Hz, (c) $N = 256$ and $df = 4$ Hz.

Frequency Resolution Impact

Importance of appropriate selection of the frequency Resolution in Spectrum Analysis

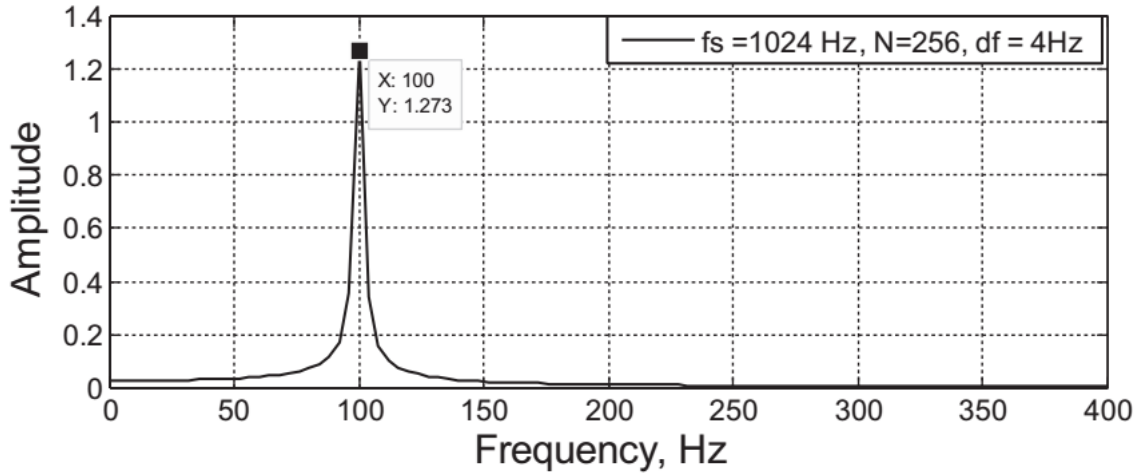


FIGURE 5.15 Estimated spectrum plot—peaks at 99 Hz and 101 Hz appearing as a peak at 100 Hz when $N = 256$ and $df = 4$ Hz used.

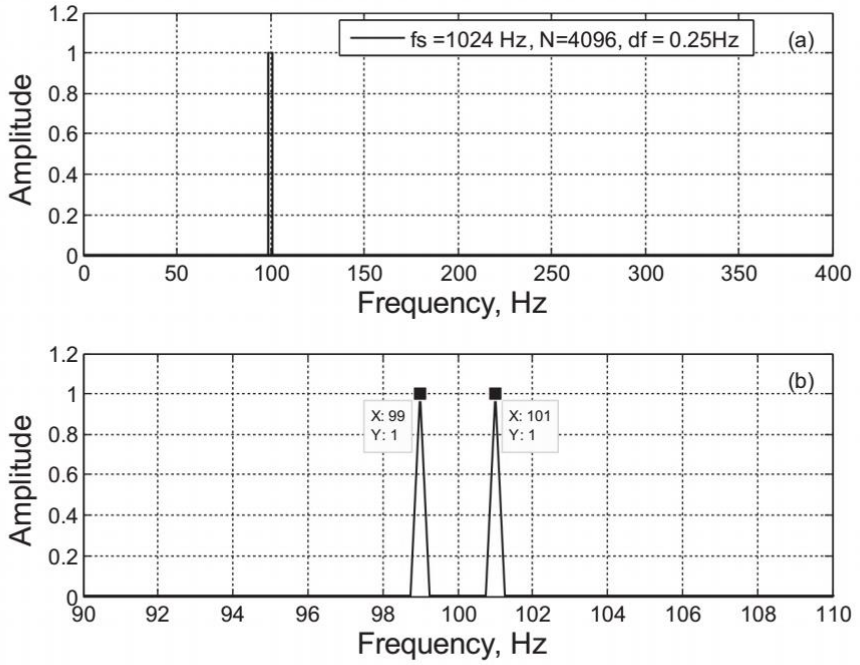


FIGURE 5.16 Estimated spectrum plots: (a) peaks at 99 Hz and 101 Hz clearly seen when $N = 2048$ and $df = 0.50$ Hz used, (b) zoomed view in frequency range 90 Hz to 110 Hz.

Leakage

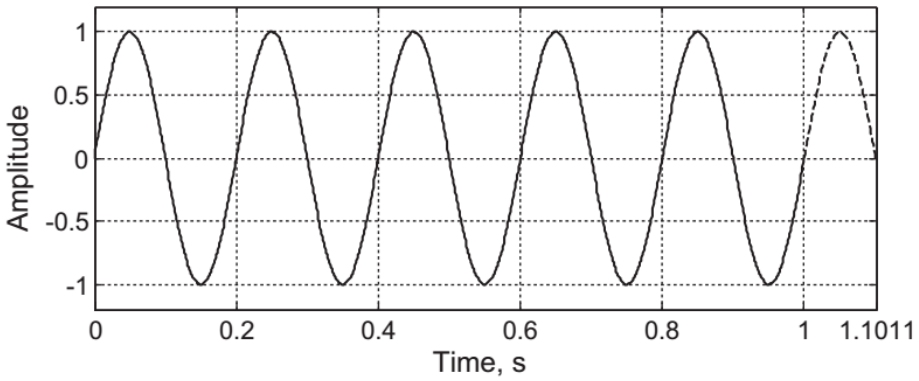


FIGURE 5.17 A typical segment size $T = N$dt of a sine waveform.

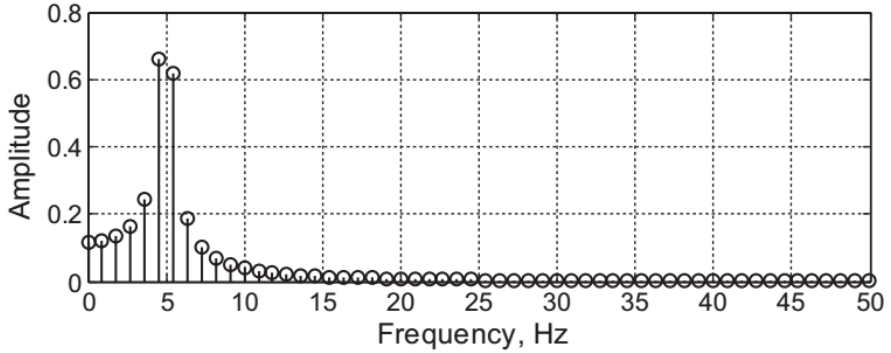


FIGURE 5.18 Frequency peak with side peaks indicate the leakage in the FT.

sine wave of 5 Hz is considered.
sampling frequency, fs = 930 Hz up to the number of the data points, N = 210 = 1024.

signal of size N = 2^{10} = 1024 of time length, T=N dt = 1.101 s at the sampling frequency, fs = 930 Hz,

The time period of the signal (Ts) is equal to 0.2 s for the 5 Hz sinusoidal signal.

But the selected segment size (N) of the segment time length, T is not the integral multiple of the time period, Ts. The extra data in the selected segment size, T = Ndt for the FT analysis is deliberately shown as the dash line from 1 to 1.1011 s in Figure 5.17. This may impact the spectrum feature calculated using Equation (5.10).

The spectrum should have a frequency peak at 5 Hz with amplitude equals to 1 for the time domain signal shown in Figure 5.17. However the artificial selection of time period, T for the segment of the signal for the FT analysis has resulted in the spectrum shown in Figure 5.18.

The spectrum is shown up to 50 Hz only to clearly observe the features around 5 Hz.
The appearance of small side peaks on either side of 5 Hz is often due to the fact that the time length of the segment may not be equals to or integral multiple of the actual time period of the signal. This effect is known as the **leakage in the spectrum**

Window Function

The use of the window function is found to overcome the leakage problem in signal processing. A few different types of windows are listed here:

1. Flat top window/rectangular window
2. Gaussian window
3. Hanning window
4. Hamming window

The idea is to make the selected segment of size, T = Ndt of any signal into periodic signal artificially, by using either of these windows to reduce the leakage problem in the spectrum using FT approach.

Window Function

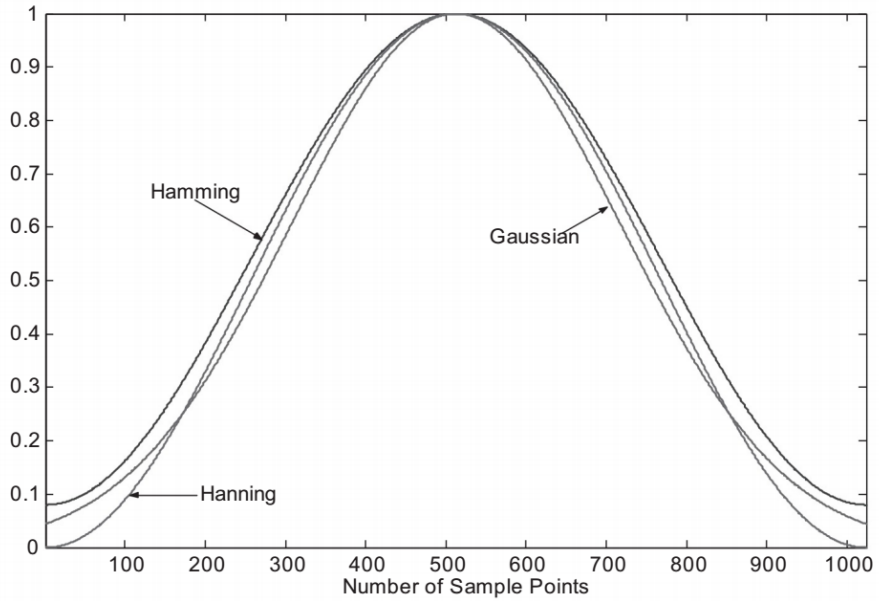


FIGURE 5.19 Typical view of a few different windows for the segment size, $N = 1024$.

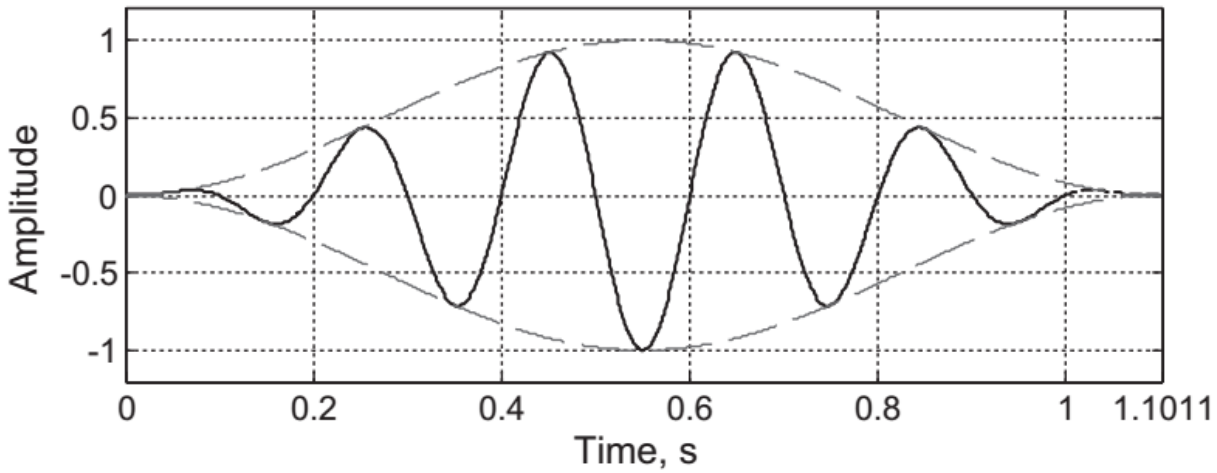


FIGURE 5.20 Sine waveform of Figure 5.17 with Hanning window.

Window Function

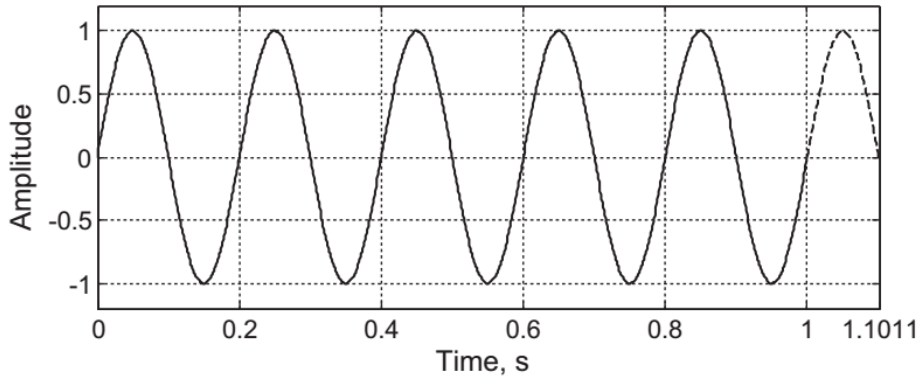


FIGURE 5.17 A typical segment size $T = N$dt of a sine waveform.

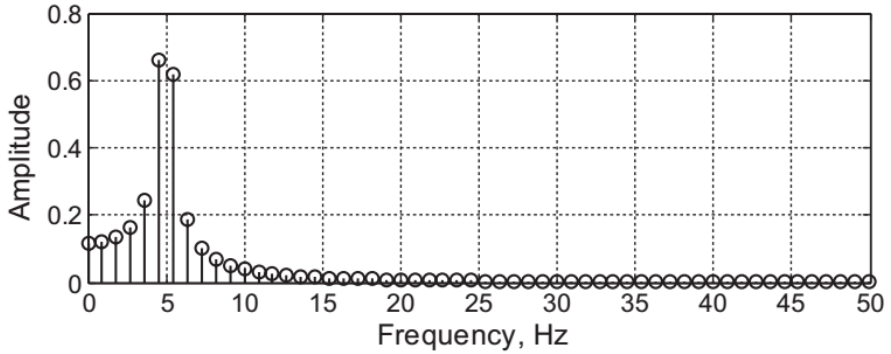
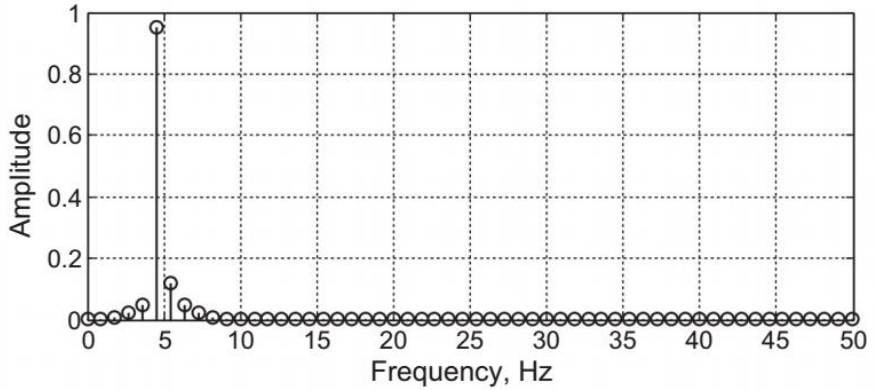


FIGURE 5.18 Frequency peak with side peaks indicate the leakage in the FT.



RE 5.21 Spectrum with reduced leakage.



Thank you

Office hours:
M-Th 3:00pm-4:30pm room C402

Please send all questions to:
gehad@example.com